

VIVA QUESTIONS

1. SS

- a.** What is forward reference
- b.** Explain assembler output format
- c.** Functions of pass 1
- d.** Functions of pass 2
- e.** Structure of SYMTAB
- f.** What is location counter
- g.** Basic loader functions
- h.** Explain algorithm for absolute loader
- i.** What is program relocation
- j.** What is bit masking in relocation loader

2. OS

- a.** Explain the difference between pre-emptive and non-pre-emptive scheduling.
- b.** How do you measure the performance of a scheduling algorithm?
What metrics are commonly used?
- c.** How does the FCFS scheduling algorithm work?
- d.** What is the "starvation problem," and how can it affect SJF scheduling?
- e.** How does the Round Robin scheduling algorithm work?
- f.** Why is Round Robin considered fair in terms of time-sharing?
- g.** How does aging help prevent starvation in priority scheduling?
- h.** Describe how the FCFS disk scheduling algorithm works
- i.** How does the SCAN disk scheduling algorithm function?
- j.** Explain how C-SCAN scheduling differs from SCAN scheduling
- k.** What is page replacement
- l.** Explain the concept of the LRU page replacement algorithm

Answers

1. SS

a. What is forward reference

A forward reference is when a program references a variable, label, or subroutine that is defined later in the code

b. Explain assembler output format

	Column	Contents
Header Record	1	H
	2-7	Program name
	8-13	Starting address of object program (HEX)
	14-19	Length of object program in bytes (HEX)
Text Record	1	T
	2-7	Starting address for object code in this record (HEX)
	8-9	Length of object code in this record in bytes (HEX)
	10-69	Object code (HEX, 2 columns per byte of object code)
End Record	1	E
	2-7	Address of first executable instruction in object program (HEX)

c. Functions of pass 1 assembler

Assign address to all statements

Save addresses of labels defined

Perform processing of assembler directive instructions

Write the intermediate file

d. Functions of pass 2 assembler

Assemble instructions

Generate data values defined by BYTE and WORD

**Perform processing of assembler directive instructions
which is not done in pass1**

Write object program and assembly listing

- e. Structure of SYMTAB

Label Name and Label Address

- f. What is location counter

**A variable used by an assembler to keep track of the current
memory address where the next instruction or data element
will be placed**

- g. Basic loader functions

Allocation, Linking, Loading, and Relocation

- h. Explain algorithm for absolute loader

See the PPT

- i. What is program relocation

**process of adjusting the addresses in a program so it can run
in any memory location rather than a fixed address. This
allows the operating system to load the program into
different memory locations**

- j. What is bit masking in relocation loader

**Bit masking in a relocation loader is used to identify which
parts of a Text record require address modification during
loading.**

1->change required, 0->nochange

2. OS

- a. Explain the difference between pre-emptive and non-pre-emptive scheduling.

In pre-emptive scheduling, the operating system can interrupt and reassign the CPU from a running process to another, allowing more responsive multitasking. In non-pre-emptive scheduling, once a process starts executing, it runs until it completes

- b. How do you measure the performance of a scheduling algorithm? What metrics are commonly used?

CPU utilization, throughput, turnaround time, waiting time, response time

- c. How does the FCFS scheduling algorithm work?

The First-Come, First-Served (FCFS) scheduling algorithm processes tasks in the order they arrive in the ready queue. When a process is added to the queue, it waits for its turn based on its arrival time; the CPU serves the first process in the queue until it completes, then moves to the next.

- d. What is the "starvation problem," and how can it affect SJF scheduling?

The "starvation problem" occurs when a process is perpetually denied the resources it needs to execute because higher-priority processes keep arriving and getting scheduled first. In Shortest Job First (SJF) scheduling, this can lead to longer processes waiting indefinitely if shorter processes continuously enter the queue, potentially causing unfairness and delaying the execution of the longer processes.

- e. How does the Round Robin scheduling algorithm work?

The Round Robin scheduling algorithm assigns a fixed time slice, or quantum, to each process in the ready queue. Processes are executed in a cyclic order, with each process getting a chance to run for a set time before being pre-empted and moved to the back of the queue if it hasn't completed

- f. Why is Round Robin considered fair in terms of time-sharing?

Round Robin is considered fair in time-sharing because it allocates equal time slices to all processes in the queue

- g. How does aging help prevent starvation in priority scheduling?

Aging is a technique used in priority scheduling to gradually increase the priority of waiting processes over time.

- h. Describe how the FCFS disk scheduling algorithm works

The FCFS (First-Come, First-Served) disk scheduling algorithm services disk requests in the order they arrive in the queue. When a request is made, it is added to the end of the queue, and the disk head moves to fulfill each request sequentially from the front of the queue until all requests are completed

- i. How does the SCAN disk scheduling algorithm function?

The SCAN disk scheduling algorithm, also known as the elevator algorithm, operates by moving the disk arm in one direction (either towards the outer or inner tracks) and servicing all requests along the way until it reaches the end of the disk. Once it reaches the end, the arm reverses direction and services requests on the return trip.

- j. Explain how C-SCAN scheduling differs from SCAN scheduling

C-SCAN (Circular SCAN) scheduling differs from SCAN scheduling in that it services requests in one direction only, moving from the innermost to the outermost track and then jumping back to the innermost track without servicing any requests during the return trip. This creates a circular motion, providing a more uniform wait time for requests

k. What is page replacement

Page replacement is the process in operating systems that occurs when a program requires more memory pages than are currently available in physical memory. When a new page needs to be loaded into memory, and there is no free space, the operating system must select one of the currently loaded pages to evict. The choice of which page to replace is determined by a page replacement algorithm, such as FIFO, LRU, or LFU, to optimize performance and minimize page faults.

j. Explain the concept of the LRU page replacement algorithm

The Least Recently Used (LRU) page replacement algorithm is based on the principle that pages that have not been used for the longest time are least likely to be needed in the near future